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COMPUTER NETWORKS

Ex. No:13	NETWORK FORENSICS USING PCAPXRAY TOOL
DATE:	

PcapXray is a **network forensics tool**. Its main purpose is to take a captured network traffic file (`.pcap` or `.pcapng`) and help you understand **what happened on the network**

AIM:

To analyze a PCAP file using PcapXray and visualize network activity for basic network forensic investigation.

ALGORITHM: For Installing PCAP and Analyzing:

1. Install PcapXray dependencies using: sudo pip3 install -r requirements.txt
2. Launch the tool with: sudo python3 PcapXray.py
3. Open the web interface and upload the given PCAP file.
4. Click Analyze to process the packet data.
5. View the generated network topology and communication graph.
6. Examine hosts, protocols, and suspicious connections.
7. Save the generated report and diagrams for documentation.

Working entirely in Chrome without installing:

https://wiki.wireshark.org/SampleCaptures

What to do

1. Open the link above.
2. Find **ipv4frags.pcap**.
3. Download it to your **Downloads** folder.
4. Open your online PCAP analyzer.
5. Click **Upload**.
6. Select:

   ipv4frags.pcap

7. Click **Analyze**.

Working entirely in Chrome without installing:

Yes. Let me show you **exactly what to click and what to look for** in A-Packets. You can do this entirely in Chrome.

Step 1 — Open A-Packets

Open the official upload page:

A-Packets PCAP Upload

You should see **"Choose a PCAP"** / **"Click or Drop to Select."** A-Packets currently accepts `.pcap`, `.pcapng`, and `.cap` files, with a **25 MB free limit**.

Step 2 — Get a sample PCAP

For your first experiment, use a **sample PCAP**, not your personal network traffic.

You can download `ipv4frags.pcap` from the Wireshark sample captures collection:

Wireshark Sample Captures

Save the file in your **Downloads** folder.

Step 3 — Upload the file

On A-Packets:

Click/Drop to Select

Then:

Downloads → ipv4frags.pcap → Open

You should see the filename appear on the page.

Then click:

Upload PCAP and Analyze

A-Packets says the analysis normally takes only a short time, depending on the capture.

Step 4 — Examine the results

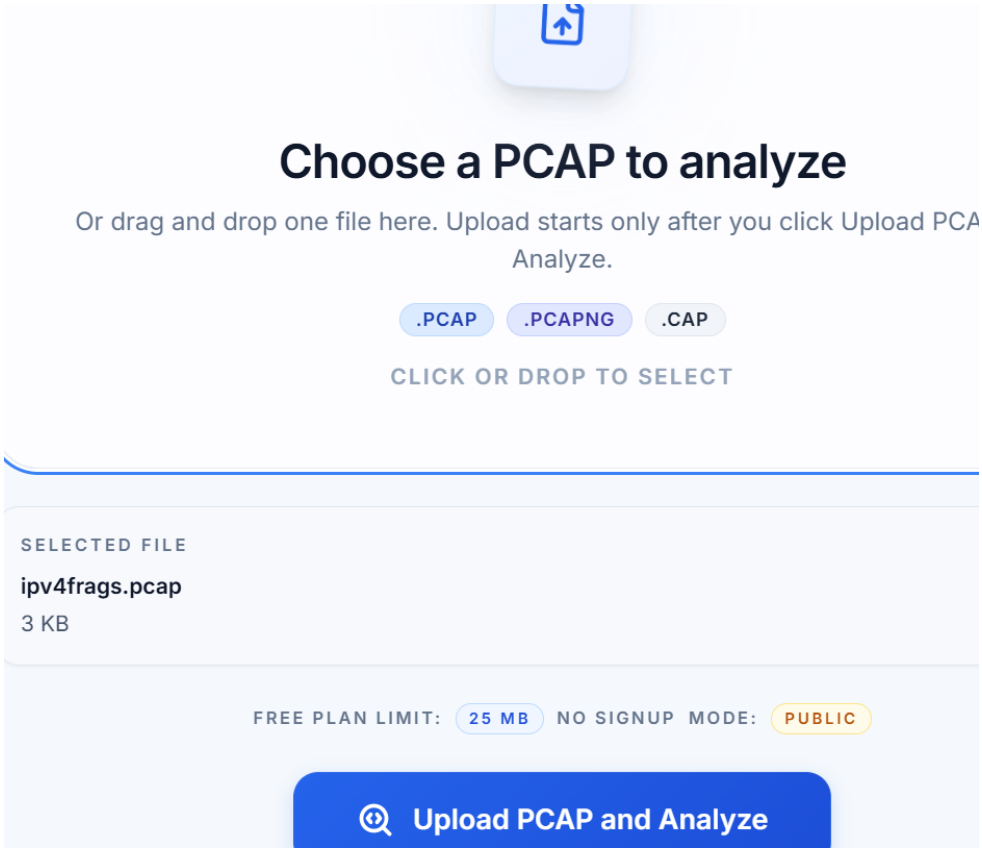
After processing, you'll get a results page.

Summary:

This gives you an overview of the PCAP, such as:

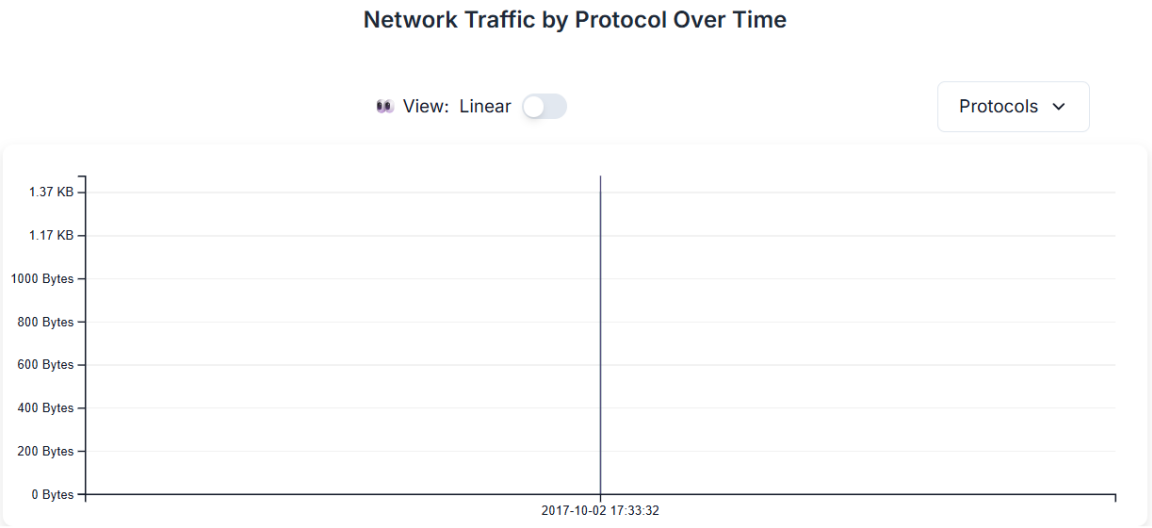
- Number of packets
- Capture duration
- Traffic information
- Hosts involved

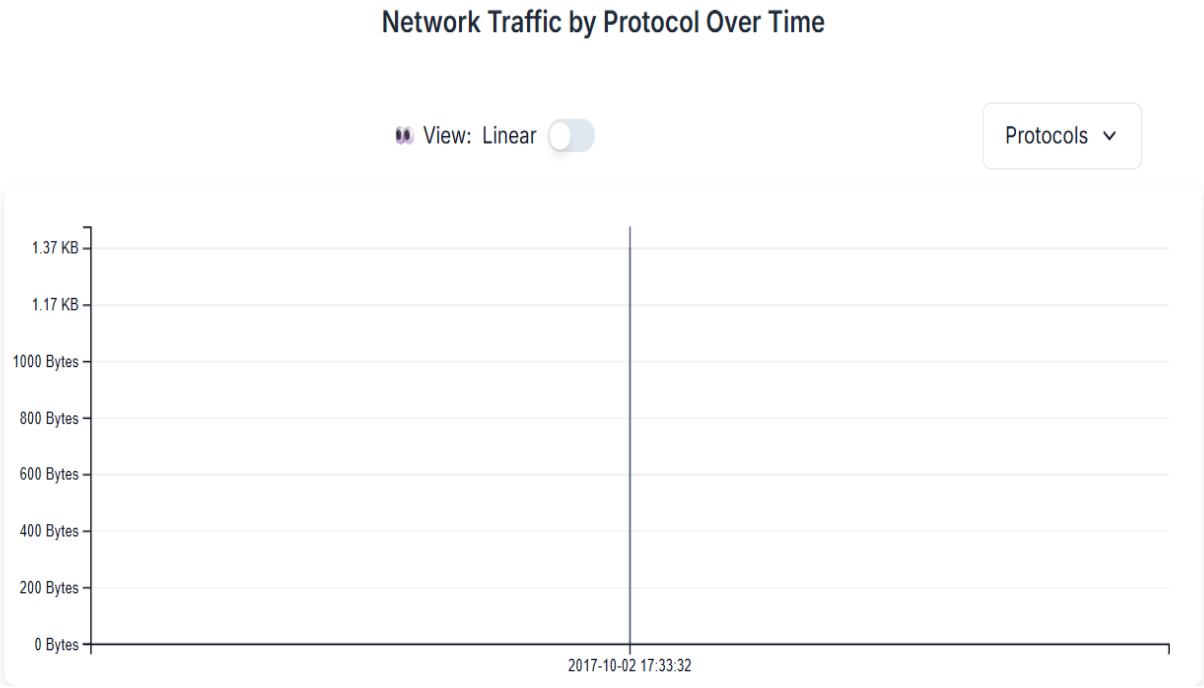
The A-Packets documentation describes a summary and traffic analysis view

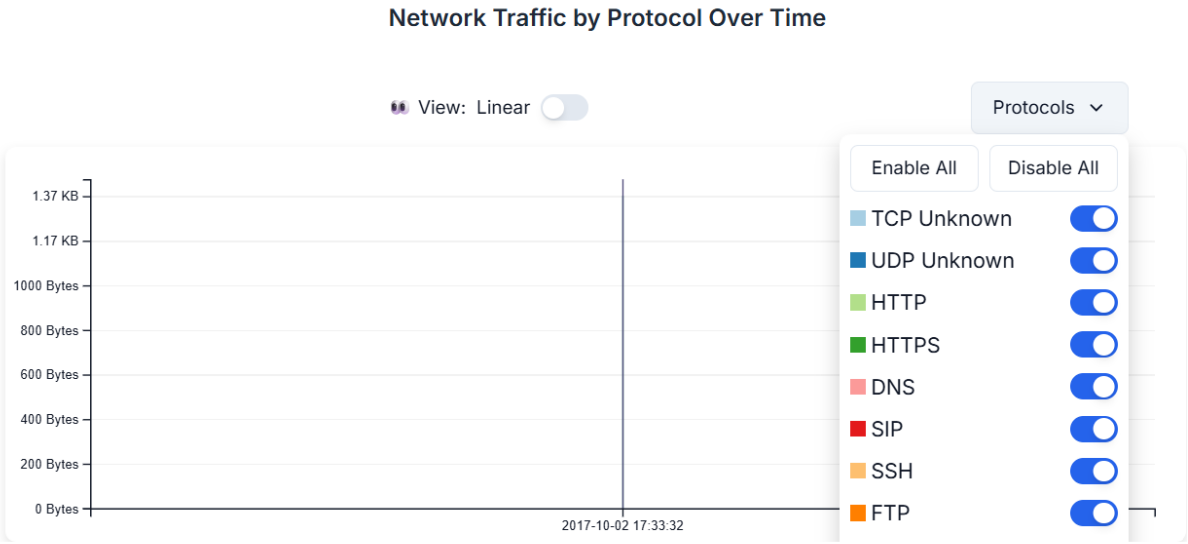


After Uploading PCAP View Report:

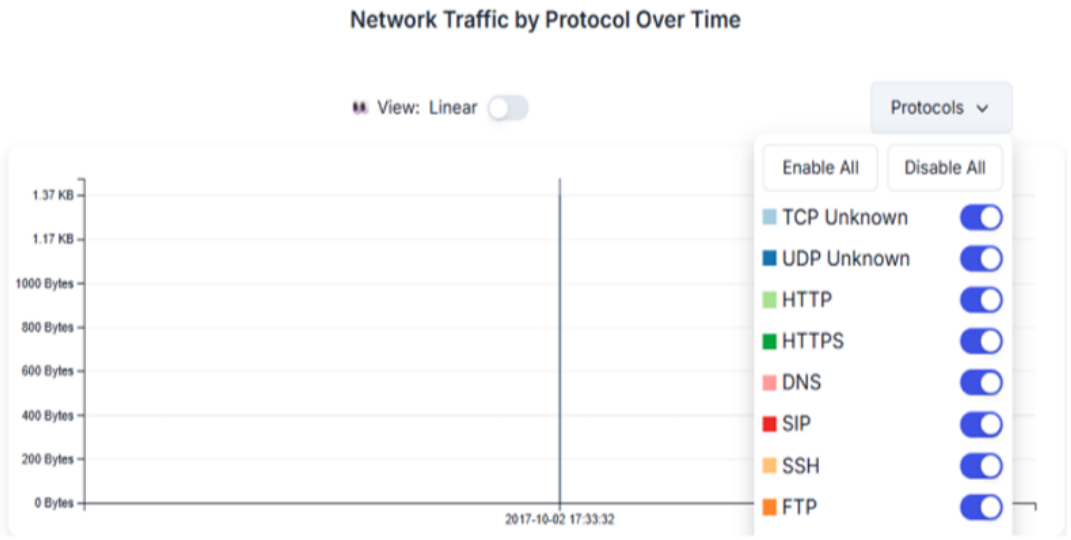
SAMPLE OUTPUTS:

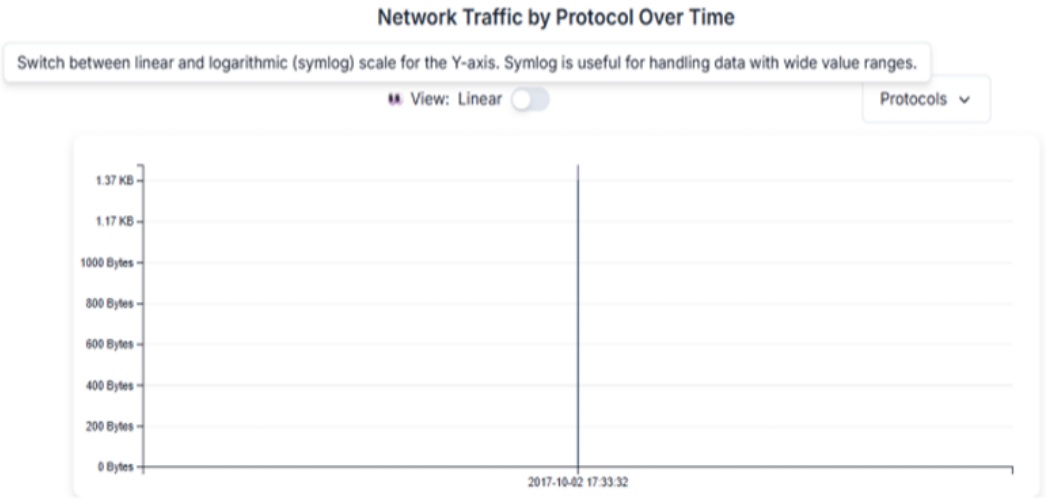





Network Traffic by Protocol Over Time
View: Linear
Protocols
Enable All
Disable All
TCP Unknown
UDP Unknown
HTTP
HTTPS
DNS
SIP
SSH
FTP
1.37 KB
1.17 KB
1000 Bytes
800 Bytes
600 Bytes
400 Bytes
200 Bytes
0 Bytes
2017-10-02 17:33:32

OUTPUT:


Network Traffic by Protocol Over Time
View: Linear
Protocols
Enable All
Disable All
TCP Unknown
UDP Unknown
HTTP
HTTPS
DNS
SIP
SSH
FTP
1.37 KB
1.17 KB
1000 Bytes
800 Bytes
600 Bytes
400 Bytes
200 Bytes
0 Bytes
2017-10-02 17:33:32


Network Traffic by Protocol Over Time
Switch between linear and logarithmic (symlog) scale for the Y-axis. Symlog is useful for handling data with wide value ranges.
View: Linear
Protocols
1.37 KB
1.17 KB
1000 Bytes
800 Bytes
600 Bytes
400 Bytes
200 Bytes
0 Bytes
2017-10-02 17:33:32

RESULT:
The PCAP file was successfully analyzed using PcapXray, and the network activities, communicating hosts, and protocols were identified and visualized.